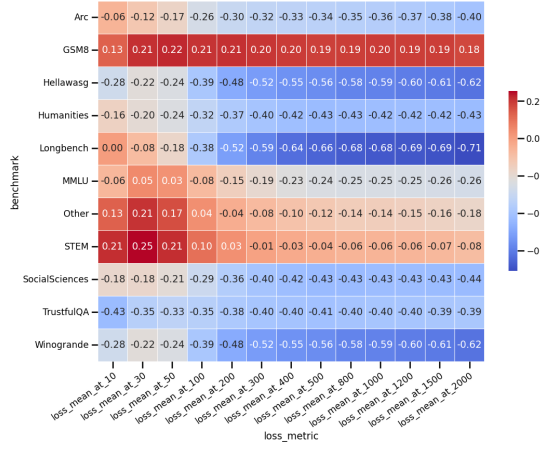
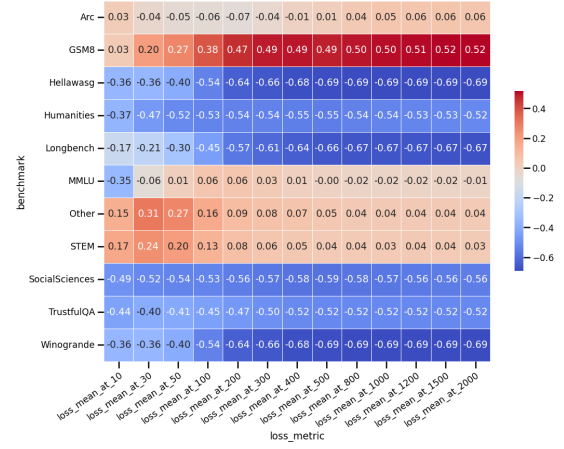


Table 10: Sample of Questions and Their Respective Answers (Correct Answer in Bold): The percentages represent the proportion of forecasters' votes on a prediction website.

Question	Start Time	End Time	Possible Answers
What will be the 12-month percentage change in the US Consumer Price Index (CPI) for September 2024?	May 31, 2024 05:00PM UTC	Oct 01, 2024 07:01AM UTC	Up by less than 1.200% or down: 0% Up by at least 1.200%, but less than 1.800%: 0% Up by at least 1.800%, but less than 2.400%: 53% Up by at least 2.400%, but less than 3.000%: 44% Up by at least 3.000%, but less than 3.600%: 3% Up at least 3.600%, but less than 4.200%: 0% Up by 4.200% or more: 0%
If Israeli Defense Forces (IDF) ground forces invade the Gaza Strip before 7 November 2023, when will Israel publicly announce or acknowledge that IDF ground forces have left the Gaza Strip?	Oct 17, 2023 05:00PM UTC	Oct 10, 2024 07:01AM UTC	Before 30 November 2023: 0% Between 30 November 2023 and 13 January 2024: 0% Between 14 January 2024 and 13 March 2024: 0% Between 14 March 2024 and 11 June 2024: 0% Between 12 June 2024 and 9 October 2024: 2% Not before 10 October 2024: 96% IDF ground forces will not invade the Gaza Strip before 7 November 2023: 2%
What will Kamala Harris' favorability rating be as of 3 October 2024, according to FiveThirtyEight?	Sep 17, 2024 08:00AM UTC	Oct 03, 2024 07:01AM UTC	Lower than 42.0%: 1% At least 42.0%, but less than 44.0%: 4% At least 44.0%, but less than 46.0%: 7% At least 46.0%, but less than 48.0%: 70% At least 48.0%, but less than 50.0%: 15% At least 50.0%, but less than 52.0%: 2% At least 52.0%, but less than 54.0%: 1% 54.0% or higher: 0%



(a) STEM: all chat models



(b) STEM: 7B level chat models

Figure 12: (a) presents the correlations by loss for all 15 chat models on STEM Academia texts. (b) presents the correlations by loss for all 9 7B level chat models on STEM Academia texts.

Table 11: Overview of AI Benchmarks and Their Evaluation Focus

Benchmark	Description
HellaSwag	Tests an AI’s common sense reasoning by requiring it to complete sentences or narratives that reflect everyday logic, highlighting the model’s ability to predict logical continuations. Zellers et al. (2019)
MMLU	Evaluates an AI’s breadth and depth of understanding across a diverse range of subjects, from science to literature, showcasing its versatility and comprehensive grasp of human language. Hendrycks et al. (2021)
TruthfulQA	Focuses on the model’s capacity to provide honest and accurate responses, particularly to questions where the truth may not be intuitive, testing its commitment to truthfulness and its proficiency in avoiding misinformation. Lin et al. (2022)
Winogrande	Assesses common sense reasoning and natural language understanding through sentence completion tasks, measuring an AI’s ability to apply common sense knowledge in language processing. Sakaguchi et al. (2019)
GSM8K	Challenges AI models with elementary-level math problems, testing their mathematical reasoning capabilities and their understanding and application of basic math concepts. Cobbe et al. (2021)

Table 12: Correlation with Different Benchmarks, Base vs. Chat Models

Benchmarks	Loss of Base Model	Loss of Chat Model	Delta
MMLU	-0.61	-0.22	0.39
MMLU:Humanities	-0.59	-0.42	0.17
MMLU:SocialSciences	-0.54	0.42	0.96
MMLU:STEM	-0.29	0.02	0.31
MMLU:Other	-0.47	-0.1	0.37
Arc	-0.46	0.34	0.8
Hellawasg	-0.66	-0.57	0.09
Wnogrande	-0.56	0.57	1.13
Longbench	-0.04	-0.65	-0.61
GSM8K	-0.4	0.22	0.62
TrustfulQA	0.05	0.39	0.34

Table 13: Correlation with Longbench values across different lengths

Length	10	30	50	100	200	300	400	500	800	1000	1200	1500	2000
Base Models	0.35	0.13	0.06	-0.03	-0.05	-0.06	-0.05	-0.05	-0.04	-0.04	-0.04	-0.04	-0.04
Chat Models	0.01	-0.07	-0.16	-0.39	-0.53	-0.58	-0.61	-0.63	-0.65	-0.65	-0.67	-0.67	-0.68